# Teaching and Examination Scheme
# I Year II Semester: B.Tech.
# (Common to all UG Programmes)

## Syllabus

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| **Name of the Programme**: B.Tech. in Computer Science & Engineering | **Year**: I | **Semester**: II |
| **Course Name:** Problem Solving using Object Oriented Paradigm | **Course Code**: | **Credit**: 2 |
| **Max Marks:** 100 | **CIE:** 40 | **SEE:** 60 |
| **End Term Exam Time:** 3 Hrs | **Teaching Scheme:** 2L+0T+0P | |

| Module No. | Contents | Hours |
|---|---|---|
| **1** | **Introduction:** Objective, Scope, Outcome of the Course and Prerequisite | 1 |
| **2** | **Object Oriented Paradigm:** Features of object-oriented programming, Difference between object oriented and procedure-oriented programming, Difference between structure and class<br>**Introduction to C++:** Data types, Operators, Conditional Statements, Loops, Arrays, Input and output streams (cin, cout), introduction to namespace. | 6 |
| **3** | **Classes and Objects**: Specifying a class, creating objects, accessing class members, defining a member function inside and outside class, access specifiers, inline function, static data members & member functions. Objects as function arguments, friend functions, returning objects to functions.<br>**Constructors and Destructors**: Need for constructors, types of constructors: default, parameterized, copy constructor, order of execution of constructors, destructors and their need. | 7 |
| **4** | **Inheritance and types of Inheritance:** Defining derived class, modes of inheritance, types of inheritance, ambiguity in inheritance, virtual base class, Function overriding, Member Classes: Nesting of Classes. | 4 |
| **5** | **Polymorphism, Pointers & Virtual Functions:** Introduction & types of polymorphism Function overloading, operator overloading, rules for overloading operators, overloading of unary & binary operators, Constructor Overloading, Friend Function. Declaring & initializing pointers, pointer to objects, this pointer, pointer to derived classes, static and dynamic binding. | 6 |
| **6** | **Files:** Introduction to File streams, Hierarchy of file stream classes, File operations, File I/O, File opening Modes, Reading/ Writing of files, Random-access to files. | 4 |
| **Total** | | **28** |